Overview of Components

Automated Barrier System: A Comprehensive Overview

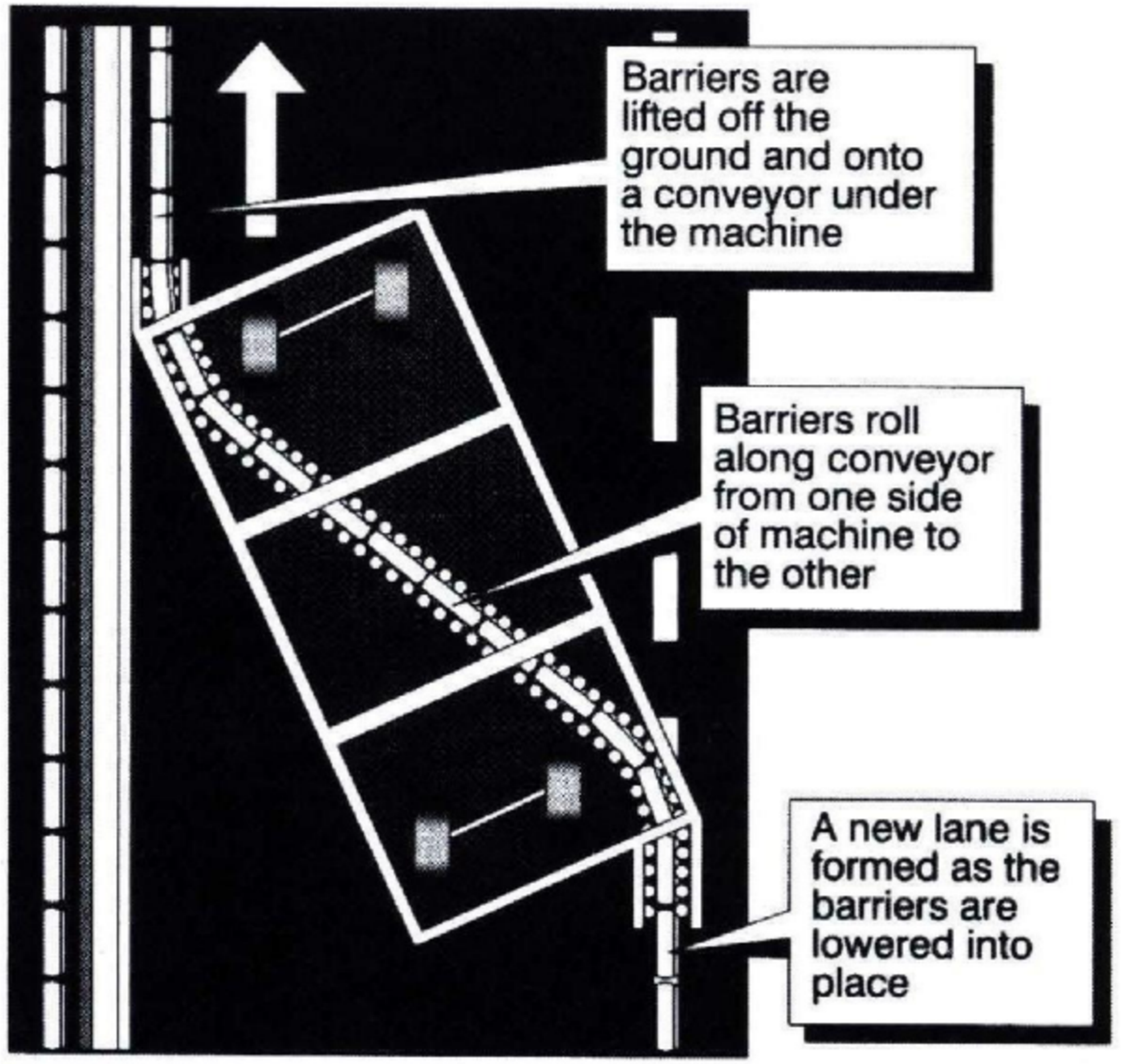


Fig 4.3 Use Automatic Guidance System For Permanent Barrier

Omnidirectional Wheels

Purpose:

- Enables side-to-side (lateral) movement of the barrier.
- Provides flexibility in tight lane spaces.

Working:

- Uses multiple rollers at angles to allow multi-directional movement.

Placement:

- Mounted underneath the base of the movable barrier.

Role in System:

- Facilitates smooth lateral lane shifts.
- Reduces turning complexity.

Understanding the Wheel Encoder

Purpose:

- Measures how far the barrier has moved.
- Provides feedback to the controller for position tracking.

Working:

- Counts wheel revolutions or pulses (incremental or absolute).
- Translates rotation to linear movement in meters.

Placement:

- Attached to motor shaft or wheel axle.

Role in System:

- Ensures accurate repositioning.
- Prevents overshoot or drift of the barrier.

Understanding the Motor Driver

Purpose:

- Controls the power supplied to motors.
- Enables speed and direction control via signals.

Working:

- Accepts control input from Raspberry Pi (PWM signals).
- Delivers appropriate current to motors.

Placement:

- Between Raspberry Pi and motors in the electronics box.

Role in System:

- Provides motive force to shift barrier.
- Should be sized based on total barrier weight and friction.

Raspberry pi (Controller Unit)

Purpose:

- Acts as the brain of the system, coordinating sensor data and motor control.

Working:

- Processes inputs from sensors (camera, encoders), executes obstacle detection algorithms, and sends control signals to actuators.

Placement:

- Mounted securely inside the main control housing of the barrier system.

Role in System:

- Runs decision-making logic
- Interfaces with camera and sensors
- Sends PWM/control signals to motor drivers
- Communicates with BMS for power status

Camera Module (Pi Camera)

Purpose:

- Detects obstacles (vehicles, people, debris).

Working:

- Streams video to Pi.
- Uses image processing (e.g., OpenCV) for object detection.

Placement:

- Mounted at the sides of the barrier chassis.

Role in System:

- Prevents collision during movement.
- Allows smarter automation.

Battery (Li-Ion or LiFePO4)

Purpose:

- Supplies power to the motor, Pi, and camera.

Working:

- Stores DC energy.
- Rechargeable via solar panel or wired supply.

Placement:

- Within the barrier body in a sealed enclosure.

Role in System:

Allows the barrier to operate without external power supply.

Solar Panel

Purpose:

- Provides energy to charge the battery during the day.

Working:

- Converts sunlight to DC electricity.
- Charges battery through charge controller/BMS.

Placement:

- Mounted on top surface of the barrier.

Role in System:

- Supports long-term autonomy and green energy use.
- Reduces need for frequent charging infrastructure.

Understanding the Battery Management System

Purpose:

- Manages charging/discharging of the battery safely.

Working:

- Monitors voltage, temperature, and current.
- Prevents overcharging or deep discharge.

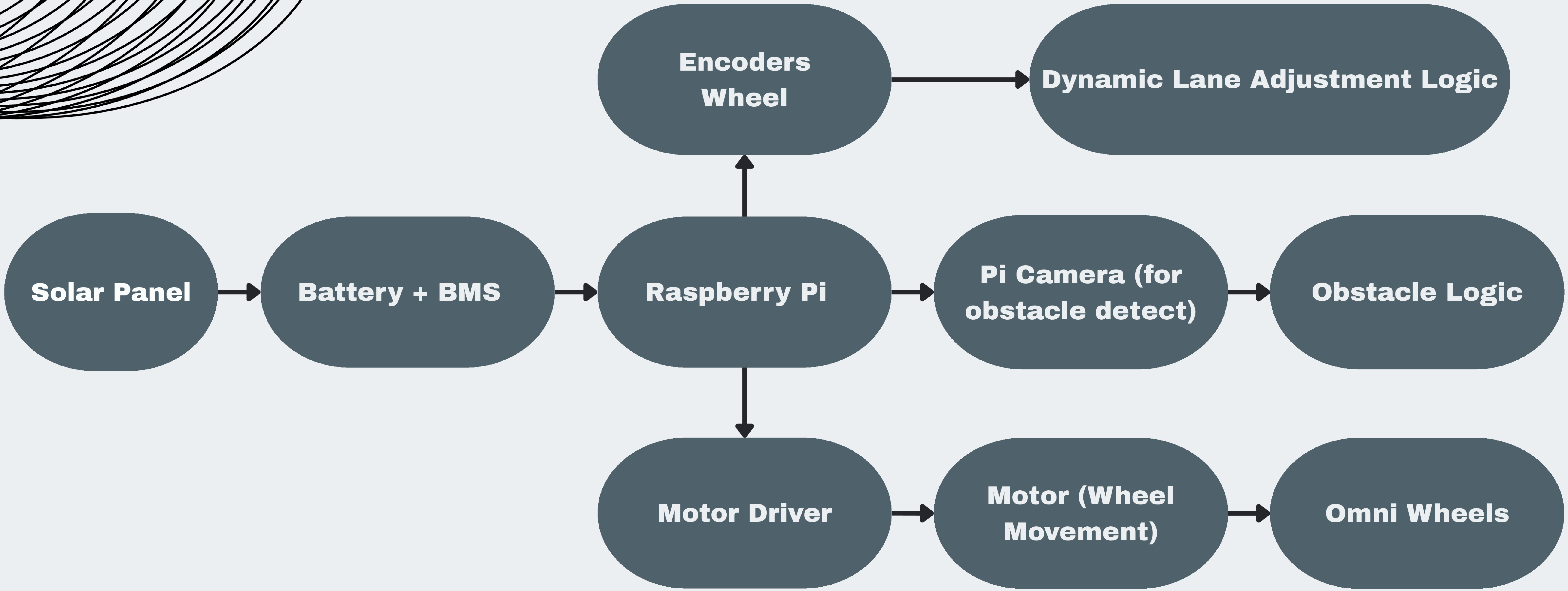
Placement:

- Integrated into the battery enclosure.

Role in System:

- Ensures long life and safety of the power system.
- Critical for autonomous operation.

System Integration Overview



Thank you for your attention!

Questions?